QND Computer Science Day 13

Mr. Schmidt

Backup plan: Arrays and Wordle

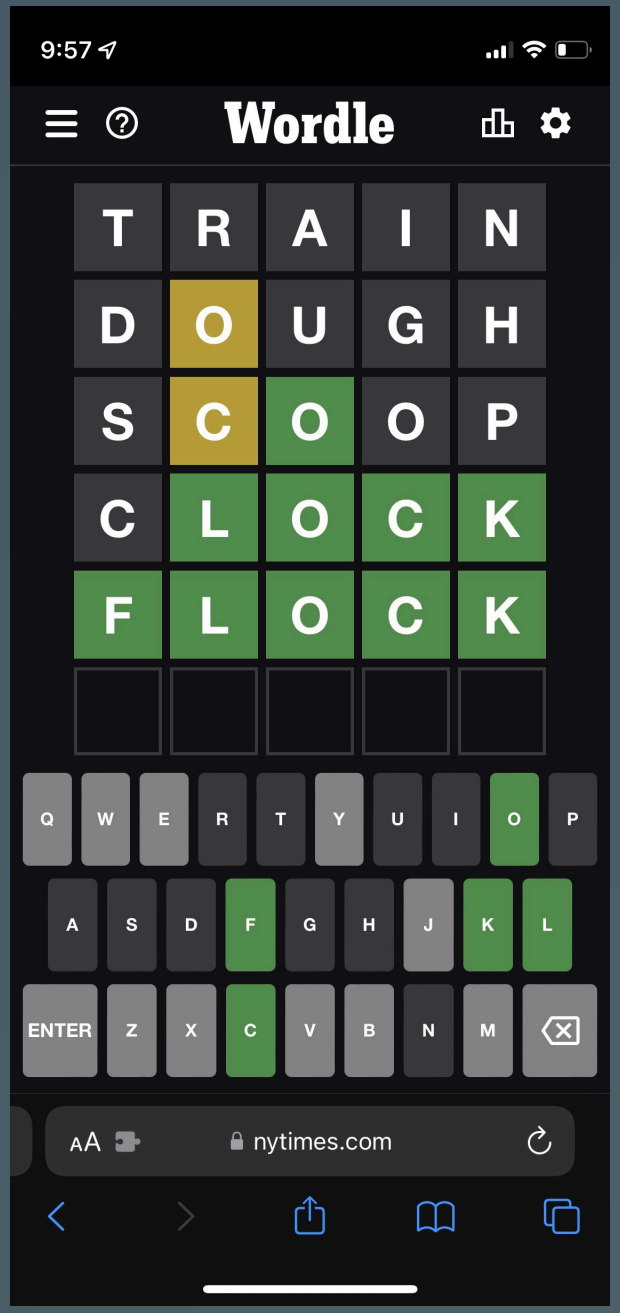
Today

- Arrays
- Wordle (Stages 1-3)
  - Use the **Text** playground

What's Next?

Wordle

- Very fun game
- Very simple



Putting a value in a String

```
let flavor = "Chocolate"

console.write("The flavor is " + flavor)
console.write("The flavor is \(flavor)") // Same thing!
```

- `\( )` puts a value inside a string
- Wordle's code uses this

What does this output?

```
let ice_cream_flavors = ["Chocolate", "Vanilla", "Cookies and Cream", "Strawberry"]

for flavor in ice_cream_flavors {
    console.write("The flavor is \(flavor)")
}
```

What does this output?

```
let ice_cream_flavors = ["Vanilla", "Chocolate", "Cookies and Cream", "Strawberry"]

let my_favorite = ice_cream_flavors[2]
console.write("My favorite flavor is \(my_favorite)")
```

Why do Arrays start at zero?

- It's a single block of memory
  - The first element is at the first address
- It's what lots of other languages do

What happens when we change 2 to 4?

```
let ice_cream_flavors = ["Vanilla", "Chocolate", "Cookies and Cream", "Strawberry"]
                         // 0          1              2                  3

let my_favorite = ice_cream_flavors[2]
console.write("My favorite flavor is \(my_favorite)")
```

Array Contains

- How to check if a Array contains an item?

```
if ice_cream_flavors.contains("Chocolate") {
    console.write("Yum!")
}
```

Wordle

- Multiple stages, each one builds on the last

Stage 1: Grade a Guess

- Turn a string into an Array
- Get a guess from the user
- Check each letter of the secret word
  - `for i in 0..<5` counts 0, 1, 2, 3, 4

Capital Letters

- `Hello` is not `hello`
- iPad capitalizing your guess? Add `.lowercased()`

```
let guess = Array(console.read("Guess a word:").lowercased())
```

Stage 2: Looping and Validation

- Put the guess part in a loop
- Check if the guess is the right length
  - Use `continue`
- Check if the guess is correct!

While Loops

- `while true` will repeat forever

```
while true {
  console.write("This will repeat forever!")
}
```

`break`

- Use `break` to exit a loop!

```
while true {
  let command = console.read("Enter a command")
  if command == "q" {
    break
  }
  // do something else
}
```

Stage 3: Guess Count Limit

- Track the number of guesses
- If guesses >= 6, tell the user they lost
- Random word from a word bank!
  - `wordBank.randomElement()!`